Mini Project Report
Prediction of Medical Insurance Charges

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Subject : Data Analytics Using Python (DAP)

1. Objective and Data Source

1.1 Objective

The major objective of the project was to conduct an in-depth analysis of the Medical Insurance Charges Dataset to identify and statistically validate the most important personal attributes affecting an individual's annual medical insurance charges. The secondary objective is to use a **Linear Regression model** to build a predictive tool that could estimate, with accuracy, a customer's cost based on identified factors.

1.2 Data Source

The **Medical Insurance Charges Dataset** was used for this project, ranking among the best resources available for regression analysis studies. After cleaning the dataset, there are **1,337 unique entries** in total with several key input features such as age, sex, bmi, children, and smoker, while charges act as the target variable.

1.3 Methods Used

Preprocessing: The project followed a structured data analysis pipeline using Python libraries such as Pandas, Seaborn, SciPy, and Scikit-learn.

- **Data Cleaning and Preparation (Step 2):** A single duplicate was eliminated. These categorical variables, including sex and smoker, were changed into numerical data with the help of the efficient .map()

approach. The region column is omitted to make the model of the baseline a focused one.

- **Exploratory Data Analysis (Step 3):** The univariate analysis established that the target variable charges is **highly skewed to the right**. Multivariate analysis was done by visualizing relationships in the form of a **correlation heatmap** and scatterplots.
- **Statistical Analysis (Step 4):** There were two independent groups, which had different variances, so the formal comparison of the charges between Smokers and the Non-Smokers was compiled using the **Independent Two-Sample T-Test**.
- **Predictive Modeling (Linear Regression) (Step 5):**Scaled the features with **StandardScaler (Z-score)** following an **80/20 train-test split** and fitted a Linear Regression model, after which the model was evaluated.

2. Key Insights and Findings

2.1 Dominant Predictors and Data Challenges

1. **Smoker Status is the Primary Driver:** The correlation heat map measured the smoker status as the only strongest predictor of cost with **high correlation of 0.79**. **Age and bmi at +0.30 and +0.20 respectively** were taken as secondary predictors.
2. **Interaction Effect Confirmed:** The multivariate scatterplot showed a critical pattern charges **increase much faster with age for smokers than for non-smokers, confirming that smoking accelerates financial risk over time**.
3. The **p-value** of the **T-test** between smokers and non-smokers was very low and it was nearly 0. This is an indication of complete statistical denial that the difference in cost is very **statistically significant** and cannot be attributed to a mere coincidence.

2.2 Model Performance and Evaluation

A Linear Regression model with good performance was embraced, which is a sign that the chosen personal attributes are very powerful predictors of cost.

1. **Model Accuracy (R^2):** The R^2 of the model was obtained to be **0.8046** that is, the model was using **80.46%** of the variance in insurance charges to be effectively explained by our chosen features.
2. **Model Error Analysis:** The **MAE of $4198.11** was the average prediction error. Nevertheless, the larger **RMSE value of $5,991.82** is a confirmation that this model is sensitive to the most extreme, high-cost outlier patients, with a significantly positive skew value of the data.

3. Conclusion and Key Takeaway

Key Takeaway: This project revealed that the visual patterns observed in the **EDA** were certainly confirmed by the statistical test (p-value about 0) and that such strong evidence transferred directly to the **final Linear Regression model**, where smoker was the leading feature in the successful prediction of more than three-quarters of variation in charges. **That confirms that individual lifestyle habits-smoking and demographics through age are the primary controllable determiners of medical insurance prices.**

4. Use of AI Tools

Name of Tool Used: Gemini 2.5 Pro

Percentage of plagiarism: 0**%** (As per Turnitin)

Percentage of AI Tool Usage: 3**%** (As per Quilbot)

Portions Where AI Was Used:

- **Language Enhancement:** Minor grammar and sentence-structure refinement for clarity and readability.
- **Code Review:** Verified correctness and readability of Python code related to regression model steps.